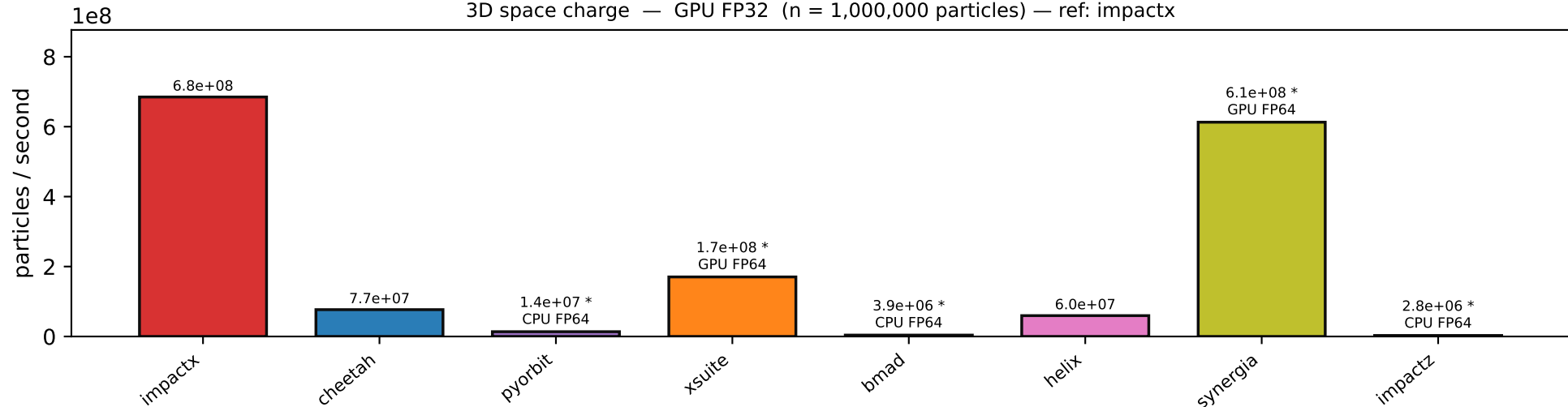


3D space charge — GPU FP32 (n = 1,000,000 particles) — ref: impactx



\* next-best shown: *pyorbit*, *bmad*, *impactz* = CPU FP64 (no GPU); *pyorbit* = exact stand-in model; *xsuite*, *synergia* = GPU FP64 (no FP32)

versions: impactx 26.08 · cheetah 0.8.2 · pyorbit 96c8def · xsuite 0.53.4 · bmad 20260904-1 · helix v1.10.1 · synergia v2024.04-209-g795b99f39 · impactz V2.7.7